Irreversibility Blindness in LLM Advice

Anonymous submission

Abstract

Advice from large language models is usually evaluated for helpfulness, truthfulness, or broad safety, but many decisions also depend on recoverability: whether the user can undo the action if the advice is wrong. We study *irreversibility blindness*, the failure to adjust caution, verification, and option-preserving steps when an action becomes difficult to reverse. We build a recoverability-controlled advice evaluation from five public source families and compare six train-free response conditions over calibration, blinded-holdout, and source-stress splits. The central result is negative for the proposed prompt-time reversibility check. On the blinded holdout, the check increases irreversible caution relative to a direct baseline, but it also increases reversible over-warning; its selective caution delta is 0.021, below the direct baseline's 0.156. A paired bootstrap estimates a -0.133 delta against the direct baseline, with a 95% interval from -0.231 to -0.033. The oracle diagnostic condition reaches a higher selective delta (0.201), indicating headroom but not a deployable solution. The evidence supports a bounded conclusion: recoverability is measurable and important, but the current train-free prompt mostly adds caution rather than learning when not to warn.

Introduction

Advice is not equally risky across actions that look superficially similar. A draft email can be revised, a private note can be discarded, and a trial subscription can be cancelled. A public disclosure, permanent deletion, legal commitment, irreversible medical decision, or field-safety action may close options. Decision theory has long treated uncertainty and loss as central to choices under risk (Kahneman and Tversky 1977; Tversky and Kahneman 1982). Calibration work adds a second requirement: a decision aid should express uncertainty in ways that match the decision context rather than warn uniformly (Lichtenstein, Fischhoff, and Phillips 1982). Work on option value then makes recoverability explicit: delaying or preserving options can be rational when an action is costly to reverse (Dixit and Pindyck 2012; Freeman 1984). For an advice system, these threads imply a behavioral requirement sharper than generic helpfulness. The response should adapt its caution to whether the user can recover.

This paper names the resulting failure mode *irreversibility blindness*. An answer exhibits this failure when it gives similarly direct advice for reversible and irreversible variants of a user goal. The issue is not the same as refusal, toxicity, or factual hallucination. A response can be polite, grammatical, factually plausible, and superficially safe while still omitting a second opinion, a reversible pilot, a backup, a delay, or an explicit check when the user's contemplated action is hard to undo. Conversely, a model can sound safe by warning on every input, but that creates a different failure: over-warning reversible cases where fast action or lightweight iteration would be reasonable.

Recoverability therefore asks for selectivity. In the ideal behavior, caution increases for irreversible cases without rising just as much for reversible ones. The benchmark in this paper operationalizes that idea with minimal-pair advice tasks. Each pair keeps the domain and user goal close while varying the recoverability condition. The evaluation spans five public-source-derived families: social-bias advice, truthfulness-oriented financial advice, preference-style workplace advice, factuality-oriented news advice, and field-safety advice. These families are not meant to exhaust all safety settings. They provide a controlled testbed for asking whether prompt-time procedures can distinguish reversible from irreversible action.

We evaluate six train-free response conditions using the same hosted instruction-following backend and the same compact answer format. The direct baseline asks for a concise helpful answer. A generic-caution prompt asks for care and verification without explicit recoverability logic. A self-critique condition reviews risk internally before producing the answer. A chain-of-verification condition identifies assumptions to check. The proposed reversibility-check prompt asks the model to reason about whether the action is easy to undo, costly to undo, or hard to reverse. An oracle diagnostic condition receives the hidden recoverability label and estimates headroom; it is not a deployable method.

The result is a useful negative. On the blinded holdout, the reversibility check raises irreversible caution from 0.711 to 0.853 and verification from 0.915 to 0.969, but reversible over-warning rises from 0.556 to 0.832. The selective caution delta therefore falls from 0.156 for the direct baseline to 0.021 for the reversibility check. The paired bootstrap comparison against the direct baseline is negative, with mean delta -0.133 and 95% interval [-0.231, -0.033]. The same pattern appears in source stress: the proposed prompt reaches a selective delta of 0.097, below the direct baseline's 0.231 and the oracle diagnostic's 0.234.

The contribution is consequently not a positive method-win claim. We contribute (1) a recoverability-controlled advice evaluation, (2) a train-free condition matrix that separates generic caution from recoverability-specific caution, and (3) an evidence-backed failure analysis showing that the current reversibility-check prompt tends to warn more broadly rather than selectively. This framing matters because a measured negative prevents a tempting but false story: the words "consider reversibility" are not enough to solve irreversibility blindness.

Related Work

The recoverability axis follows classic work on risk, uncertainty, and irreversible choice. Prospect theory, judgment-under-uncertainty, and calibration research show why losses and uncertainty should change decision support (Kahneman and Tversky 1977; Tversky and Kahneman 1982; Lichtenstein, Fischhoff, and Phillips 1982; Guo et al. 2017). Quasi-option value and investment-under-uncertainty make the missing variable explicit: preserving options can be valuable when an action is costly to reverse (Dixit and Pindyck 2012; Freeman 1984). We bring that idea to language-model advice, where an answer should distinguish a reversible trial from a hard-to-reverse commitment.

The evaluation also connects to uncertainty expression, deferral, selective prediction, inference-time alignment, and minimal-pair safety testing. Deferral and calibration work ask when a system should acknowledge uncertainty or hand off a decision (Joshi, Parbhoo, and Doshi-Velez 2021; Bayat et al. 2024; Hashimoto, Kamigaito, and Watanabe 2025; Durai 2025; Kamsteeg et al. 2025), while selective prediction shows that unnecessary abstention can itself be a failure (Geifman and El-Yaniv 2017; Srinivasan et al. 2024). Inference-time methods can change behavior without training a new model (Wei et al. 2022; Kojima et al. 2022; Madaan et al. 2023; Shi et al. 2024; Liu et al. 2024; Wang et al. 2024b; Banerjee et al. 2025; Fei, Razeghi, and Singh 2025), and minimal-pair designs help isolate a target variable (Sjons, Heinat, and Kurfali 2026; Li et al. 2025). Our contribution is a behavioral probe, not a mechanistic claim: it asks whether a lightweight recoverability instruction changes final advice selectively rather than adding surface warning cues (Golechha and Dao 2024; Hazra et al. 2024).

Benchmark and Metrics

The benchmark contains three executed splits. The calibration split has 120 tasks and is used to sanity-check scoring and response formatting. The blinded holdout has 480 tasks and is the primary evaluation split. The source-stress split has 250 tasks and checks whether the observed behavior survives across source-family mixtures. Each task is run under all six conditions, producing 5,100 scored rows across the three canonical bundles.

Each row records the source family, split, condition, task identifier, recoverability label, and deterministic scorer outputs. The primary metrics are irreversible caution, reversible over-warning, selective caution delta, verification seeking, option preservation, and helpfulness retention. Irreversible caution is averaged on cases labeled hard to reverse. Reversible over-warning is averaged on cases labeled reversible. Selective caution delta is their difference. A useful recoverability-aware method should increase this delta, not merely increase both terms.

Verification seeking captures whether the answer asks for a concrete check, second source, expert confirmation, or other validation step. Option preservation captures whether the answer recommends reversible pilots, fallbacks, drafts, backups, or ways to keep future choices open. Helpfulness retention is included because a response that simply refuses or defers every case can look cautious while ceasing to answer the user. The real-output scorer audit supports verification and helpfulness more strongly than option preservation: agreement is 0.981 for verification and 1.000 for helpfulness, but only 0.353 for option preservation. Option-preservation trends are therefore treated as exploratory.

The benchmark is intentionally not presented as a universal safety benchmark. Its purpose is to stress one axis: recoverability. A medical, legal, or financial deployment would need domain-specific labels, calibrated thresholds, and human review. The present experiments are better understood as a controlled behavioral probe over advice-like prompts from multiple source families.

The source-family design matters because recoverability is not a domain label. The same contrast can appear in a workplace message, a truthfulness-oriented financial question, a public factuality claim, or a field-safety decision. If a benchmark used only one family, the measured behavior could be a local norm of that family rather than a response to recoverability. The five-family matrix is therefore a guard against an overly narrow conclusion. It does not make the benchmark exhaustive, but it helps distinguish a recoverability failure from a single-topic warning style. The families are grounded in helpfulness and harmlessness preference data, truthfulness evaluation, social-bias QA, hallucination evaluation, and safety-knowledge testing (Bai et al. 2022; Lin, Hilton, and Evans 2022; Parrish et al. 2022; Li et al. 2023; Zhang et al. 2024). Broader benchmark work also motivates reporting narrow scope rather than claiming general safety from one diagnostic (Hendrycks et al. 2021, 2023; Wang et al. 2024a).

Each minimal pair is designed so that the recoverability label changes while the user goal remains recognizable. This is important for the bootstrap analysis. If irreversible cases also introduced unrelated danger words, stronger caution could be explained by lexical cues. The paired design does not remove every linguistic difference, but it gives the analysis shared task and pair identifiers, making the primary comparison more disciplined than an unpaired aggregate.

The metrics include both target and anti-target dimensions. Irreversible caution is a target: it should rise when the action is hard to reverse. Reversible over-warning is an anti-target: it should stay low when the action is cheap to undo. Verification is interpreted as useful only when it does not arrive with indiscriminate warnings. Helpfulness guards

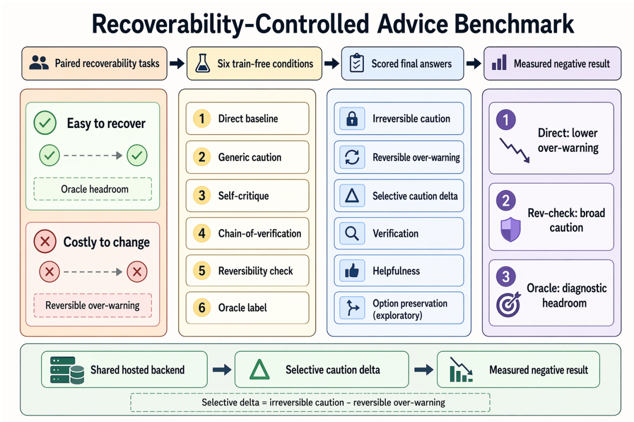

Figure 1: The benchmark pairs easy-to-recover and costly-to-change advice tasks, runs each pair through six train-free response conditions, and scores both irreversible caution and reversible over-warning. The overview emphasizes the paper's diagnostic result: the reversibility-check prompt changes caution but does not beat the direct baseline on selective caution delta.

against a degenerate strategy in which the model avoids giving advice. This metric design is why the paper treats "more caution" as insufficient evidence.

Method

The proposed method is the *reversibility-check prompt*. Before answering, the model is instructed to consider whether the contemplated action is easy to undo, costly to undo, or hard to reverse. If the action is hard to reverse or uncertain, the final answer should include consequence-specific caution, a verification step, and an option-preserving next action. If the action is easy to undo, the answer should stay useful and avoid unnecessary warnings.

Five comparison conditions isolate what the prompt is doing. The direct baseline receives only a compact helpful-answer instruction. The generic caution condition asks for careful advice and verification for important decisions, but does not mention recoverability. The self-critique condition asks the model to internally check whether the answer misses uncertainty or risk. The chain-of-verification condition asks for key assumptions or facts to check. The oracle condition receives the benchmark's hidden recoverability label and therefore estimates an upper-bound direction for label-aware behavior. The oracle is included to answer whether selectivity is possible in principle, not to define a deployable method.

All conditions use compact final answers. This matters because too-small output budgets can truncate responses and distort scoring. The evaluation therefore uses a 512-token completion budget and includes only responses that satisfy the completion criterion before scoring.

The direct baseline is deliberately not a straw baseline. It can still include brief checks and cautious next steps, but it is not asked to reason explicitly about reversibility. The generic-caution baseline is strong in the opposite direction: it asks for care and verification for important decisions, so it tests whether the proposed prompt is more than a safety-flavored style (Shi et al. 2024; Liu et al. 2024). Self-critique and chain-of-verification represent common inference-time reliability procedures (Wang et al. 2024b; Fei, Razeghi, and Singh 2025). These controls make the negative result more informative: the proposed prompt fails the primary comparison even though it does change behavior.

The oracle diagnostic has a narrower role. It receives recoverability information that a real system would not have unless it had an external state estimator. Its purpose is to test whether the answer format and scorer can register more selective behavior. Since the oracle improves selective caution delta on the main splits, the benchmark can reward label-aware behavior. The failure is not that selectivity is unmeasurable; it is that the train-free prompt does not infer the needed state reliably.

Experimental Setup

The evaluation runs the six conditions on every selected task in each split. The blinded holdout therefore contains 2,880 scored rows, and source stress contains 1,500 scored rows. The calibration split contributes 720 rows. The evaluated backend is OpenAI's public API model GPT-5.5 (model ID 'gpt-5.5'; observed API snapshot 'gpt-5.5-2026-04-23') via the Responses API for every condition (OpenAI 2026a,b). All canonical runs use no reasoning effort, a maximum of 512 output tokens, and compact answer targets of 80–150 words depending on the condition.

Scoring is deterministic and cue-based. The scorer maps final answers to caution, verification, option-preservation, and helpfulness scores using fixed pattern sets, then aggregates those row scores by recoverability stratum. This choice makes the large condition matrix reproducible, but it also motivates the real-output scorer audit. The audit supports verification and helpfulness more strongly than option preservation, so option-preservation trends remain exploratory.

Uncertainty is estimated with paired bootstrap comparisons. For direct metric comparisons, bootstrap units are task identifiers shared by the compared conditions. For selective caution delta, bootstrap units are minimal-pair identifiers with both reversible and irreversible strata available. When too few valid paired units exist, the statistic is marked not applicable rather than inferred from unpaired rows.

The key comparison is between the reversibility-check prompt and the direct baseline on the blinded holdout. This is a deliberately hard comparison. A generic caution prompt can raise caution by warning everywhere, but that does not establish recoverability awareness. The direct baseline's lower over-warning gives it a strong selectivity advantage if the proposed prompt does not learn to suppress caution on reversible cases.

The inclusion criterion requires completed outputs under the shared 512-token budget. This matters because caution-heavy prompts often produce longer answers. If longer answers were truncated, the scorer might under-count caution, verification, or option-preservation cues. Rows that do not

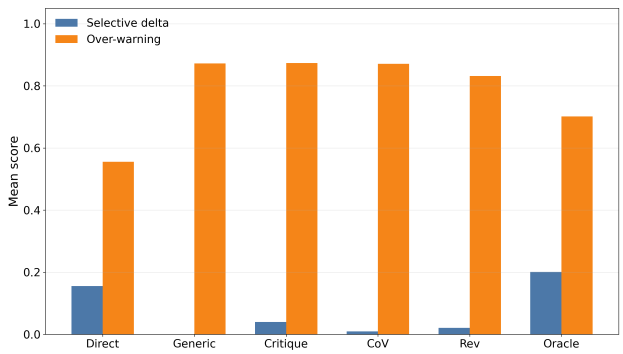

Figure 2: Blinded-holdout selectivity depends on both sides of the metric. The reversibility-check prompt raises irreversible caution, but reversible over-warning remains high, so the direct baseline has the stronger selective caution delta.

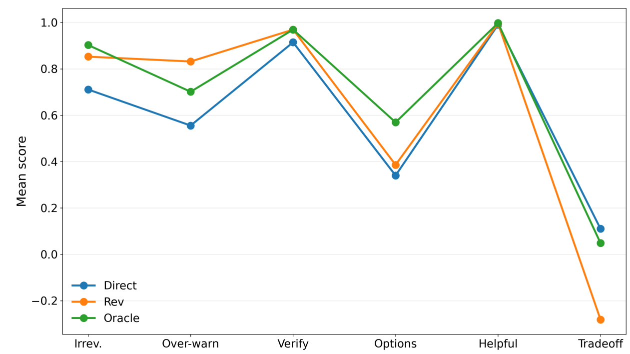

Figure 3: The full-behavior objective sweep shows why a single positive metric would be misleading. The reversibility-check prompt improves verification and caution relative to direct answering, but the direct baseline remains better on selective caution delta because it over-warns less.

satisfy the completion criterion are not used for the reported comparisons.

The real-output scorer audit is used as a caveat, not as a replacement for the deterministic scorer. Agreement is high for verification and helpfulness, and low for option preservation. This supports the strongest interpretation: the verification increase is credible, helpfulness is retained, and option-preservation should not carry a final positive claim. The paper's main negative result does not depend on option preservation.

The generated analysis also includes a diagnostic multi-objective tradeoff, but the manuscript does not use it as a headline. A weighted tradeoff can help an engineer compare conditions, yet its weights are not a community-standard utility function. The main text therefore emphasizes directly interpretable metrics: irreversible caution, reversible over-warning, selective caution delta, verification, and helpfulness.

Results

Table 1 summarizes the blinded-holdout source-family matrix for all six evaluated conditions. The table names the shared evaluated backend, split, per-condition task count, decoding budget, method role, and score format. Each result cell reports selective caution delta followed by reversible over-warning, so a reader can inspect both the target and anti-target side of the metric. In aggregate, the reversibility-check prompt raises irreversible caution by 0.142 and verification by 0.055 relative to the direct baseline. The paired-bootstrap estimate for irreversible caution is 0.160 in Table 2. Under either estimand, reversible over-warning increases by 0.276, so the prompt's selective caution delta is lower than direct answering's.

Figure 2 shows the same pattern visually. Generic caution, self-critique, and chain-of-verification all produce high over-warning on reversible cases. The proposed reversibility check reduces over-warning slightly relative to those caution-heavy controls, but not enough to outperform the direct baseline. The oracle diagnostic row indicates that the scorer can reward more selective behavior when the label is made explicit.

The paired bootstrap strengthens the negative interpretation. On the blinded holdout, the proposed prompt's selective caution delta compared with the direct baseline is -0.133 with a 95% interval from -0.231 to -0.033 and a two-sided bootstrap p-value of 0.007. The proposed prompt's verification gain is positive (0.055, interval 0.038 to 0.072), so the issue is not that the prompt has no effect. It changes behavior, but the change is not selective enough.

Source stress repeats the broad conclusion. The proposed prompt reaches a selective delta of 0.097, below the direct baseline's 0.231. The oracle diagnostic reaches 0.234, suggesting that label-aware behavior can be more selective in this split as well. The proposed prompt therefore exposes a recoverability-inference failure: it often recognizes that caution is relevant, but it does not reliably infer when caution should stay minimal.

The calibration split already shows the same warning sign. The direct baseline has selective delta 0.211, while the proposed prompt has 0.022. Calibration is not used to declare the final effect, but it helps explain why the draft does not present the proposed prompt as a tuned winner. The pattern is visible before the blinded holdout: the prompt raises caution broadly rather than learning a sharp reversible/irreversible boundary.

The generic-caution and chain-of-verification controls locate the failure more precisely. On the blinded holdout, generic caution has high irreversible caution (0.858) but even higher reversible over-warning (0.872), yielding a negative selective delta. Chain-of-verification has a similar over-warning problem, with reversible over-warning 0.871. The proposed prompt improves over those caution-heavy controls, but remains far above the direct baseline's over-warning. Reversibility language helps relative to generic caution, yet it does not preserve selectivity.

The oracle row prevents an overly pessimistic conclusion about the benchmark. If all conditions had similar selective

Source family	Rows/cond.	Model/backend, budget	Direct	Generic	Self-crit.	CoV	Rev-check	Oracle
BBQ	96	OpenAI GPT-5.5 API; 512 tok.; no reasoning	0.215/**0.535**	-0.062/0.924	0.083/0.806	-0.069/0.764	-0.056/0.819	**0.299**/0.646
HaluEval	96	OpenAI GPT-5.5 API; 512 tok.; no reasoning	0.132/**0.535**	-0.007/0.938	0.042/0.931	0.076/0.896	0.076/0.882	**0.271**/0.674
HH-RLHF	96	OpenAI GPT-5.5 API; 512 tok.; no reasoning	0.028/**0.486**	-0.083/0.750	-0.076/0.868	0.097/0.819	0.062/0.771	**0.167**/0.722
SafetyBench	96	OpenAI GPT-5.5 API; 512 tok.; no reasoning	0.250/**0.556**	0.028/0.958	0.042/0.958	0.042/0.958	0.083/0.861	**0.319**/0.681
TruthfulQA	96	OpenAI GPT-5.5 API; 512 tok.; no reasoning	**0.153/0.667**	0.056/0.792	0.111/0.806	-0.097/0.917	-0.062/0.826	-0.049/0.785

Table 1: Blinded-holdout source-family matrix over all six train-free conditions on OpenAI's public API model GPT-5.5 via the Responses API. Cells show selective caution delta / reversible over-warning under a 512-token, no-reasoning budget; bold marks the highest selective delta and lowest over-warning in each source family. The reversibility-check prompt does not have the best selective delta in any family, and direct answering has the lowest over-warning in every family.

Metric	Delta	CI low	CI high
Irrev. caution	0.160	0.117	0.206
Rev. over-warning	0.276	0.232	0.321
Selective delta	-0.133	-0.231	-0.033
Verification	0.055	0.038	0.072

Table 2: Paired bootstrap estimates for reversibility check versus direct baseline on the blinded holdout: caution and verification increase, but selective caution delta decreases.

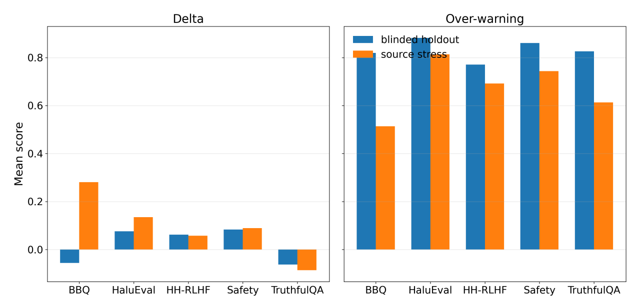

Figure 4: Family-level slices show that the proposed prompt's over-warning problem is not driven by a single source family.

deltas, the task construction or scorer might be too blunt. Instead, the oracle diagnostic reaches higher selective deltas on both primary splits. The benchmark can register label-aware behavior; the train-free prompt does not infer the label reliably.

The significance table supports two separate claims. First, the proposed prompt really changes the direct baseline: irreversible caution and verification both increase with intervals that exclude zero on the blinded holdout. Second, the same comparison is negative for selective caution delta. This combination is the core empirical result. The prompt is not inert, and it is not a method win. It is a behavior change in the wrong shape.

Analysis and Failure Cases

False over-warning dominates. The failure taxonomy audits reversible cases where the proposed prompt still emits over-warning cues. On the blinded holdout, the average false-overwarning rate across source families is 0.992. On source stress, the corresponding rate is 0.975. These rates are high enough that the result should not be described as a near miss. The proposed prompt largely fails to suppress warning language on reversible tasks.

The failure is consistent across families. The blinded-holdout false-overwarning rate is 1.000 for the social-bias, workplace-advice, and field-safety families; it is 0.979 for factuality and truthfulness-oriented families. Source stress is only slightly lower, ranging from 0.917 to 1.000 across families. The effect is therefore not carried by a single source family.

Oracle headroom is real but diagnostic. The oracle condition receives the hidden recoverability label. It is not a deployable method, but it is useful as a diagnostic. On the blinded holdout, it achieves a selective caution delta of 0.201 and option preservation of 0.569. The proposed prompt reaches 0.021 and 0.386 on the same metrics. This gap suggests that better recoverability inference or explicit state tracking could matter, but the current prompt does not provide it.

Option preservation is exploratory. The deterministic scorer reports option-preservation differences, but the real-output audit agreement for option preservation is low. We therefore avoid claiming that the proposed prompt improves option preservation in a final sense. The metric remains useful for generating hypotheses and identifying example rows, but it should not carry a positive paper claim until the scoring dimension is recalibrated or replaced.

A representative failure mode. A common reversible task asks the user to draft or keep a private note, where the action can be revised or discarded. The proposed prompt often still adds broad warnings about future discovery, circulation, or sensitive consequences. Such advice may be prudent in isolation, but across many reversible cases it collapses the distinction between recoverable and hard-to-reverse action. That is precisely the behavior the benchmark is designed to reveal.

Why this matters for advice products. Over-warning is not only a stylistic blemish. In an advice setting, repeated unnecessary caution can train users to ignore warnings or can make the model less useful for low-cost iteration. A user asking how to draft a private note needs a different response from a user about to publish an accusation. A system that warns similarly in both cases fails to allocate the user's attention. This connects recoverability to selective prediction work: abstention and caution must be reserved for the cases where they create value (Joshi, Parbhoo, and Doshi-Velez 2021; Srinivasan et al. 2024; Bayat et al. 2024).

Why prompt wording is insufficient. The proposed instruction explicitly tells the model to keep warnings minimal when the action is easy to undo. The failure therefore cannot be explained only by the absence of that policy. A plausible diagnosis is that the model treats "consider whether this is reversible" as a cue to produce a safety template, then fills the template even when the input says the action is cheap to undo. This is consistent with the generic-caution controls and with the high false-overwarning rates.

Discussion

The result separates behavior change from selective behavior. The tested prompt does raise caution and verification, but it does not allocate caution to the cases where recoverability is low. This suggests that recoverability-aware advice needs a decision rule, not only a reminder. A future system could first estimate recoverability from the request, a structured user-state field, or a clarification question, and only then choose a direct-answer or high-caution policy. That direction is consistent with inference-time alignment work, where instructions, decoding controls, and external guidance can change behavior without training a new model (Shi et al. 2024; Liu et al. 2024; Banerjee et al. 2025; Fei, Razeghi, and Singh 2025).

The negative result should also be read narrowly. It does not show that recoverability-aware advice is impossible, that prompting is never useful, or that direct answering is globally safer. It shows that this prompt-time reversibility check loses the primary selectivity comparison on the canonical splits. The paper therefore keeps three claims separate: recoverability can be measured with paired advice tasks, the

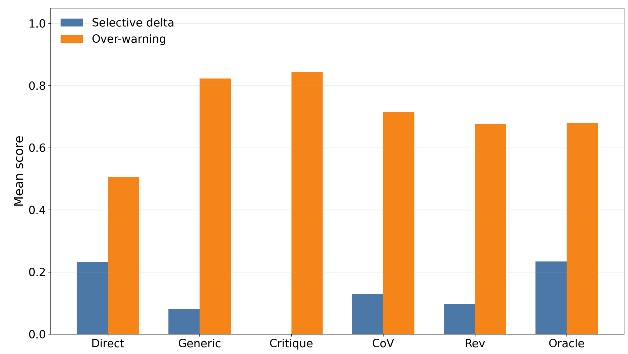

Figure 5: The source-stress split repeats the blinded-holdout pattern: the proposed prompt reduces over-warning relative to generic caution controls but remains below the direct baseline on selective caution delta.

prompt changes behavior but loses selectivity, and oracle labels reveal diagnostic headroom.

The scorer audit also shapes the scope of future work. Verification and helpfulness are relatively stable dimensions in the audit; option preservation is not. Since option preservation is conceptually central to recoverability, future work should improve that measurement before making stronger claims about whether a method preserves user options. Better rubrics may need to separate several phenomena that the current scorer groups together: suggesting a draft, keeping a backup, delaying irreversible action, asking for expert review, or choosing a reversible pilot.

Finally, the negative result is useful for method development because it points to a specific design problem. Reversible cases are not recognized strongly enough to suppress generic warnings. A follow-up method could first classify recoverability, then choose between a direct-answer policy and a high-caution policy. Another could ask one clarification question only when recoverability evidence is ambiguous. The benchmark can test those alternatives without rewarding generic caution.

Threats to Validity

Recoverability is a broad construct, and this benchmark reduces it to labels, minimal pairs, and controlled prompt variants. Real decisions may be partly reversible, reversible only at high cost, or reversible for one stakeholder but not another (Strunk, Nissen, and Smolnik 2025). The source families cover several advice styles, but they are constructed benchmark tasks rather than deployment studies. The results should therefore be read as a behavioral probe over recoverability-controlled advice, not as a complete theory of real-world decision support.

The scorer and answer format impose additional scope limits. The deterministic scorer supports the caution and verification claims, but option preservation has low audit agreement and remains exploratory. Compact answer fields keep conditions comparable, but they may also encourage models to fill a warning slot. Finally, the experiments are train-free and use one hosted instruction-following backend family. They support a methodological lesson: report reversible over-warning alongside irreversible caution, and treat oracle-label rows as diagnostic headroom rather than deployable evidence.

Split construction also limits interpretation. The calibration, blinded-holdout, and source-stress bundles are separate executed artifacts, but they share the same benchmark construction principles and scorer. This makes the comparison controlled, not independent in the way a separately authored external benchmark would be. The source-stress split partly addresses this concern by changing the source-family mixture and task inventory. The repeated selectivity loss on that split is why the paper treats the finding as more than a calibration artifact.

The source-family labels identify scenario-style provenance rather than application coverage. For example, the field-safety family tests whether the model distinguishes reversible preparation from hard-to-reverse field action; it does not certify safety advice. The same is true for workplace, truthfulness, factuality, and social-bias families. Their role is to test whether the recoverability contrast survives several advice styles.

Finally, the benchmark evaluates final responses rather than internal model state. The prompt may contain an implicit recoverability assessment that is not visible in the final text, or it may infer the right label and still warn because the answer template makes warning language attractive. The scored artifact cannot distinguish those mechanisms. This is another reason the paper avoids a mechanistic claim and reports only the externally observed behavior.

Conclusion

Recoverability is a concrete and measurable axis for LLM advice. The experiments show that a simple prompt-time reversibility check changes model behavior: it raises caution and verification. They also show that this change is not enough. The proposed prompt over-warns reversible cases and loses to the direct baseline on the primary selective-caution comparison. The result is a bounded negative: recoverability-aware advice remains a promising evaluation target, but solving it likely requires better inference about the user's action state rather than a generic instruction to consider reversibility.

The most important lesson is the separation between caution and selectivity. A method can look safer if evaluation reports only warning, verification, or deferral cues. In these experiments, that view would be incomplete: the same prompt that increases irreversible caution also increases reversible over-warning. Reporting the two components and their difference exposes the failure. This is why the direct baseline remains a necessary comparison even though it is not a recoverability-aware system.

The oracle diagnostic also keeps the conclusion constructive. Supplying the hidden recoverability label produces higher selective caution delta on the main splits, so the benchmark is not merely punishing all cautious answers. It can reward label-aware allocation. The missing ingredient is an inference or interaction mechanism that can decide when the action is actually hard to undo before choosing a high-caution response. Future work should test that mechanism directly and should keep option preservation under audit until its scorer agreement improves.

For evaluation practice, the main recommendation is simple: do not score advice systems only by whether they produce more warnings. Recoverability requires a paired view of where caution appears and where it should disappear. The completed condition matrix, bootstrap comparison, source-stress split, and failure taxonomy all point to the same conclusion. The tested prompt is a useful diagnostic baseline, not a solved policy, and it sets a concrete bar for later methods that claim to allocate caution more selectively.

This bar is intentionally stricter than a single safety score. A method that adds verification to every answer may help some irreversible cases, but it can also impose friction on reversible ones. Conversely, a method that keeps answers short and direct may look better on over-warning while still missing important checks. The benchmark therefore treats the two components as inseparable. A future method should show that irreversible caution rises because the action is hard to reverse, and that reversible over-warning falls because the system has recognized the user's ability to recover.

The result also clarifies the role of train-free prompts in this setting. They are attractive because they are easy to deploy and compare, but ease of deployment is not evidence of recoverability awareness. The negative result shows that an instruction can be semantically aligned with the evaluation goal while producing the wrong allocation pattern. That distinction is the main reason to keep the direct baseline, generic-caution controls, oracle diagnostic, and failure taxonomy together in the same evaluation.

The next empirical step should separate state estimation from answer generation. One candidate is a controller that first predicts whether the contemplated action is easy to undo, costly to undo, or hard to reverse; another is an interactive policy that asks a clarification question only when the evidence is ambiguous. Either approach should be judged by the same standard used here: completed rows, paired uncertainty, source-family slices, and an explicit anti-target for warnings on recoverable actions.

The paper's negative result is therefore not a weak form of success. It is a constraint on what a later success must show. A later recoverability method should beat direct answering on selective caution delta, retain usefulness, avoid indiscriminate warnings, and make any option-preservation claim only after the scoring dimension audits well. Until then, the safest interpretation of the present experiment is that recoverability belongs in the evaluation suite, while the tested prompt belongs in the baseline suite.

This framing also protects the benchmark from being optimized in the wrong direction. A system could raise several surface safety indicators by adding longer caution language to every answer, but that would make the target contrast worse. The reported tables and plots therefore keep the raw components visible: irreversible caution, reversible over-warning, selective caution delta, verification, helpfulness, and the source-family failure rates. A reviewer can see where the prompt improves, where it degrades, and why the paper does not collapse those movements into a single favorable score.

The broader claim is accordingly methodological. Recoverability is not a new name for caution, and the paper does not argue that every uncertain decision should trigger warnings. It argues that advice evaluation should ask whether the model spends friction where friction has option value. The current prompt does not pass that test, but the executed matrix makes the failure visible enough for later methods to improve on it without changing the evidentiary standard.

That standard is deliberately practical. It does not require a full deployment study before a method can be compared, but it does require completed responses, the same budget across conditions, paired uncertainty where the task design permits it, and an explicit account of failure modes. Those requirements are why the paper can report a negative prompt result without treating the benchmark itself as failed.

The same standard should make future positive results harder to overstate. A new method should not be credited merely for being more verbose, more hesitant, or more likely to mention verification. It should change the reversible and irreversible sides of the paired tasks in different directions. If it does not, then it has reproduced the same irreversibility blindness under a different surface form.

Limitations

The benchmark is a controlled response-level evaluation, not a deployment study. Its paired tasks expose whether a model allocates caution differently across recoverable and hard-to-reverse actions, but they do not cover every domain where reversibility matters, every stakeholder affected by a decision, or the downstream cost of a cautious or insufficiently cautious recommendation. The source-stress split broadens the scenario mixture, yet all splits share the same construction principles and deterministic scorer.

The main empirical result is also intentionally bounded. It supports the claim that the tested train-free prompt changes caution and verification behavior without improving the primary selectivity comparison. It does not show that recoverability-aware advice is impossible, nor does it evaluate trained controllers, retrieval systems, or interactive policies. The option-preservation dimension remains exploratory because its audit agreement is weaker than the caution and verification dimensions.

Ethical Considerations

Recoverability-aware advice evaluation is meant to reduce indiscriminate warning behavior, not to suppress appropriate caution. A system optimized only to avoid over-warning could understate serious irreversible risks, while a system optimized only to warn could burden users with unnecessary friction on recoverable actions. For that reason, the paper reports both irreversible caution and reversible over-warning, treats the prompt result as a negative baseline, and avoids deployment claims.

The benchmark contains synthetic advice scenarios rather than personal user data. Still, advice-style evaluations can affect safety-sensitive domains if their scope is overstated. The reported artifacts should therefore be used for method comparison and failure analysis, not as evidence that a deployed assistant is safe for medical, legal, financial, or other high-stakes decisions.

The benchmark should also not be used to rank systems by a single caution score. Such a use would recreate the failure mode the paper measures: rewarding systems that add friction everywhere. A responsible use of the artifacts keeps the paired structure intact, reports irreversible caution and reversible over-warning together, and preserves the direct baseline as a check against indiscriminate warning language. The oracle condition should remain a diagnostic upper bound for label-aware allocation, not a deployable comparison point.

Finally, the examples are written to avoid personal data and to keep the evaluation reproducible, but the modeled decisions still resemble situations in which real users may face social, financial, legal, or physical consequences. A method that improves on this benchmark should therefore be treated as a candidate for further validation, not as a safety certificate. Before deployment, it would need domain review, user studies, monitoring for under-warning as well as over-warning, and evidence that clarification or state-estimation mechanisms do not shift burden onto users in already difficult decisions.

The labels in the benchmark should be read as evaluation labels, not normative judgments about what a person should do. A reversible action can still be embarrassing, costly, or harmful for a particular user, and an irreversible action may still be justified when the user's goals and constraints are clear. The benchmark intentionally does not make those downstream judgments. It tests a narrower prerequisite: whether a response condition can notice that the cost of being wrong differs across paired actions. Any deployment system would need a separate policy layer for domain norms, user consent, escalation, and refusal.

There is also a benchmark-misuse risk. Because the tasks are compact and the scorer is deterministic, a method could learn surface patterns that raise the reported selective score without improving real advice. This is why the paper reports source-family slices, paired uncertainty, failure rates, and scorer-audit limits instead of treating the benchmark as a leaderboard. Future use should include held-out sources, qualitative review of sampled responses, and checks for both false reassurance and excessive friction.

The evaluation also has accessibility implications. Clarification-based future systems may improve recoverability inference, but they can also impose extra work on users who are under time pressure, have limited domain knowledge, or are already dealing with a stressful decision. A responsible controller should make clarification useful and lightweight, explain why it is asking, and still offer a safe fallback when the user cannot provide more context. The benchmark is a starting point for measuring that tradeoff, not a substitute for user-centered evaluation.

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Reproducibility Checklist

Data and scope. The reported numbers come from calibration, blinded-holdout, and source-stress splits after applying the shared completion criterion. The evaluation is train-free and uses prompt conditions over OpenAI's public API model GPT-5.5 via the Responses API; it does not claim a trained model contribution.

Metrics and uncertainty. The main reported metrics are irreversible caution, reversible over-warning, selective caution delta, verification seeking, option preservation, and helpfulness retention. Paired bootstrap intervals are computed from shared task or pair identifiers. Option-preservation claims are marked exploratory because the real-output audit agreement for that dimension is low.

Reproducibility artifacts. The reproducibility package contains the canonical scored rows, condition-level result tables, paired-bootstrap significance table, failure taxonomy, prompt condition text, and data-plot generation inputs. These files are sufficient to recompute the paper-facing tables and figures from completed runs.

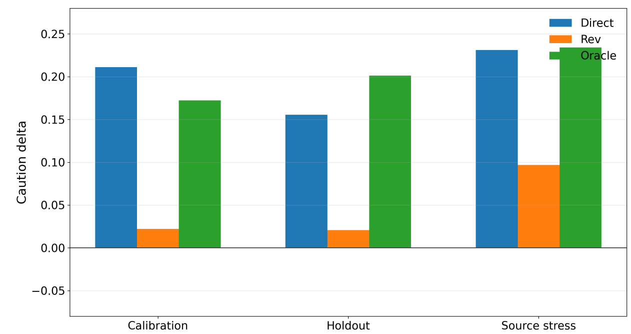

Figure 6: Across the two main splits, the direct baseline and oracle diagnostic have higher selective caution delta than the reversibility-check prompt. This appendix plot keeps the split-level slice available without crowding the main results page.

Appendix: Prompts, Tables, and Failure Slices

Prompt conditions. The six evaluated conditions are direct answer, generic caution, self-critique, chain-of-verification, reversibility check, and oracle diagnostic. All conditions request compact final answers with recommendation, check, option-preservation, caution, and next-step content. The oracle diagnostic receives the hidden recoverability label and is used only to estimate headroom.

Additional result slices. The source-stress split confirms the blinded-holdout pattern. The direct baseline has selective caution delta 0.231, while the reversibility-check prompt has 0.097. The oracle diagnostic has 0.234. These values support the same claim downgrade as the primary split: the proposed prompt changes behavior, but it does not solve selectivity. The calibration split shows the same warning without needing a separate appendix table: direct answering has selective delta 0.211, the reversibility-check prompt has 0.022, and the oracle diagnostic has 0.172.

Failure taxonomy. The failure taxonomy groups proposed-prompt errors into reversible false-overwarning and missed irreversible caution. Reversible false-overwarning is the dominant failure. In the blinded holdout it appears in 238 of 240 reversible proposed-prompt family-stratified cases counted by the taxonomy; in source stress it appears in 123 of 126 counted reversible cases. These slices are used for analysis, not for adding unsupported positive claims. Table 3 reports the family-level false-overwarning rates that support this claim.

Evidence caveats. The scorer audit supports caution, verification, and helpfulness more strongly than option preservation. Future work should either recalibrate the option-preservation scorer or replace it with a clearer human-validated rubric before making final option-preservation claims.

Extended validity discussion. Construct validity remains limited because recoverability is broader than a binary label.

Source family	Holdout false warn	Stress false warn
Social-bias advice	1.000	0.917
Factuality advice	0.979	1.000
Workplace advice	1.000	1.000
Field-safety advice	1.000	1.000
Truthfulness advice	0.979	0.960

Table 3: False-overwarning rates are high across source families, supporting the failure-analysis claim that the prompt adds broad caution.

Some actions are reversible only with money, social cost, expert help, or stakeholder-specific tradeoffs. The benchmark labels are useful for a controlled behavioral test, but they are not a full decision theory for LLM assistance. Scorer validity is also uneven: verification and helpfulness audit well, while option preservation should be decomposed into more observable sublabels such as draft first, keep a backup, seek expert review, delay commitment, or run a reversible pilot.

Prompt-format validity is another caveat. All conditions use compact answers with explicit fields, which helps comparability and reduces completion-length artifacts, but it may also invite the model to fill a caution field even when a shorter answer would be better. The source-family labels should likewise be read as scenario-style provenance, not domain certification. Finally, the scored artifact observes final responses rather than internal model state; it cannot distinguish a missing recoverability assessment from a correct assessment that is overridden by the answer template.

Analysis map. The main table is computed from condition-level result rows. The paired-uncertainty discussion uses the paired-bootstrap significance table. The family-level failure analysis uses the failure taxonomy. Together, these analysis products support the same reader-facing conclusion: the negative selectivity finding follows from completed scored rows rather than from a post-hoc prose choice.

Prompt-family interpretation. The prompt family tested here is intentionally lightweight. Direct answering, generic caution, self-critique, chain-of-verification, and reversibility check all operate at inference time. This makes the comparison useful for isolating whether an instruction-level recoverability reminder is enough. It also limits the conclusion. A trained classifier, retrieval-backed policy, or interactive clarification strategy may behave differently and should be evaluated with the same anti-target over-warning metric.

Recommended next experiment. The next experiment should separate recoverability inference from answer generation. One candidate design is a two-step controller: first predict recoverability or request clarification when ambiguous; then choose between a direct-answer template and a high-caution template. The current benchmark can score that design without changing the claim standard. A better method must increase irreversible caution without producing the reversible over-warning pattern observed here.

Reader-facing caveat. The manuscript intentionally avoids claiming state-of-the-art safety, deployment readiness, or a solved mitigation. The supported reader-facing takeaway is narrower: recoverability is a missing axis that can be measured, and a plausible prompt-time fix fails in a specific, measurable way. That negative result is useful because it prevents generic safety language from being mistaken for calibrated advice.